Statistical Arbitrage in Cryptocurrencies

Momentum and Reversal Research on Live Market Data - Version 2

Wall Street Quants - Bootcamp Project

Survivorship-free universe of 39 coins | data 2021-01-01 to 2026-10-08

Revised after trainer feedback, October 2026

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Executive Summary

This is version 2 of the project. It answers the class brief - research momentum and reversal strategies in crypto, test them with unconstrained backtests at 20 bps (market orders) and 7 bps (limit orders), combine them with a weighting scheme and report returns, volatility, Sharpe, drawdown and alpha / beta - and it fixes the five points raised in the trainer's review of version 1 (Section 0).

Headline: the hold-out result

All research and every design decision used only 2021-07 to 2023-12. On the untouched hold-out (2024-01-01 to 2026-10-08) the combined momentum + reversal portfolio earned a net Sharpe of 0.20, an annualised alpha of 2.8% with a t-statistic of only 0.52, a beta to Bitcoin of -0.03 and a correlation to Bitcoin of -0.12. In the research period the same portfolio had a Sharpe of 0.93 and an alpha t-statistic of 1.64. The edge did not survive out of sample at a statistically meaningful level.

Table 1. Combined portfolio, net of costs. Hold-out first.

Combined portfolio	Sharpe	Ann. return	Ann. vol	Max DD	Alpha	Alpha t	Beta	Corr BTC
Hold-out 2024-2026	**0.20**	**2.1%**	**10.3%**	**-26.7%**	**2.8%**	**0.52**	**-0.03**	**-0.12**
Research 2021-2023	0.93	9.4%	10.1%	-13.3%	9.6%	1.64	-0.03	-0.15
Full backtest	0.50	5.2%	10.2%	-26.7%	5.7%	1.42	-0.03	-0.13
Buy & hold BTC, hold-out	0.61	24.1%	39.4%	-53.1%	0.0%	0.33	1.00	1.00

** Momentum book at 20 bps (market orders), reversal book at 7 bps (limit orders). Sharpe = mean / std x sqrt(252); return, volatility and alpha annualised with 252 days. Alpha, beta and correlation vs daily Bitcoin returns; alpha t uses Newey-West (5-lag) errors.*

Key findings

- **Horizon.** With 39 coins chosen by market cap at the start of 2021, research-period data show cross-sectional reversal at 1-5 days (1-day gross Sharpe -0.85) and at 21-120 days, and time-series momentum at 1-4 week horizons (0.45 at 30 days).
- **Costs.** Short-horizon reversal is real in gross terms but its turnover makes it lose money even with 7 bps limit orders; only slow signals survive the brief's costs.
- **What survived.** Time-series momentum with an implied-volatility (DVOL) filter is the one component that held up: research net Sharpe 0.55, hold-out 0.62 at 20 bps. Theme momentum and the slow reversal signals that looked best in 2021-2023 decayed.
- **Version 1 vs version 2.** Version 1 reported a Sharpe of 1.02 using sqrt(365). On the class sqrt(252) scale that is 0.85 (full) and 0.46 (hold-out). Removing survivorship bias and hold-out contamination takes the hold-out Sharpe from 0.46 to 0.20.

0. Changes in Version 2 (Response to Trainer Feedback)

Table 2. Trainer feedback and the version 2 response.

Trainer comment	Change in version 2	Effect
Sharpe used sqrt(365); class uses mean / std x sqrt(252)	All Sharpe ratios use sqrt(252); returns, volatility and alpha are annualised with 252 days (sa_core.ANN = 252)	V1 combined Sharpe 1.02 -> 0.85 (full), 0.55 -> 0.46 (hold-out)
Hold-out not untouched: exploration ran on 2021-2026 and the 60/90-day reversal only appears in 2024-2026	Pre-registered mechanical protocol: every signal's direction, every candidate, the execution rule and the weighting are chosen on 2021-07 to 2023-12 only (Section 4.4)	The hold-out is now a true out-of-sample test; the combined hold-out Sharpe is 0.20
Lead with hold-out numbers; alpha t falls from 2.19 to 0.90; add	Hold-out reported first everywhere (summary, tables, README, deck); correlation to BTC added to every	Hold-out alpha t = 0.52; correlation to BTC = -0.12

correlation to BTC	table	
Universe: 30+ coins by market cap at the start, not the 16 survivors	Top 40 coins by market cap on 2021-01-03 (stablecoins and wrapped tokens excluded); 39 have free data. Coins that collapsed or were delisted (FTT, EOS, MKR) are kept until their prices end	Removes survivorship bias; the universe includes the FTX collapse and other failures
Be able to explain every choice and rebuild the backtest	src/minimal_backtest.py rebuilds a strategy in ~60 lines and checks it against the engine; a separate methodology explainer covers every choice and likely interview questions	Engine and minimal backtest agree to machine precision

1. Introduction

Statistical arbitrage looks for price-volume patterns that predict returns. The brief highlights momentum (recent winners keep winning) and reversal (recent moves undo themselves), and gives a research outline for each: time horizon, new information and activity, seasonality, investment themes and technical plays for momentum; time horizon, uninformed trading, correlation and macro dislocation for reversal. It then asks for unconstrained backtests, realistic execution costs, a weighting scheme and standard performance metrics.

Version 2 keeps the research outline of version 1 but changes how conclusions are drawn. The universe is fixed by market cap at the start of the sample, so it includes coins that later failed. All exploration and selection uses the 2021-2023 research period, through a mechanical rule set written down before the hold-out is examined. The research question is unchanged: which momentum and reversal patterns exist in crypto, which survive realistic costs, and how should they be combined?

2. Literature Review

Cross-sectional momentum (Jegadeesh & Titman, 1993), time-series momentum (Moskowitz, Ooi & Pedersen, 2012) and industry momentum (Moskowitz & Grinblatt, 1999) are robust across asset classes, but momentum crashes in high-volatility rebounds (Daniel & Moskowitz, 2016) and benefits from volatility scaling (Barroso & Santa-Clara, 2015). Short-term reversal (Lehmann, 1990) is a reward for supplying liquidity: low-volume moves reverse more (Campbell, Grossman & Wang, 1993) and reversal profits rise with market volatility (Nagel, 2012). Long-horizon reversal reflects overreaction (De Bondt & Thaler, 1985). In crypto, Liu and Tsyvinski (2021) and Liu, Tsyvinski and Wu (2022) document a momentum factor, Han, Kang and Ryu (2024) find time-series momentum more robust than cross-sectional momentum after costs, and Fil and Kristoufek (2020) find pairs relationships unstable. Garleanu and Pedersen (2013) motivate partial trading toward targets; DeMiguel, Garlappi and Uppal (2009) show that simple weights are hard to beat. Harvey, Liu and Zhu (2016) and Bailey and Lopez de Prado (2014) warn that testing many strategies inflates backtest Sharpe ratios - the reason for version 2's protocol. Survivorship bias, the reason for version 2's universe, is a standard concern in any backtest that conditions on assets that still exist at the end of the sample.

3. Data

3.1 A survivorship-free universe

Version 1 traded 16 coins that were still liquid on Coinbase in 2026, which quietly excludes every coin that failed. Version 2 fixes the universe at the start of the sample: the 40 largest coins by market capitalisation in the CoinMarketCap historical snapshot of 2021-01-03, after removing stablecoins (USDT, USDC, DAI, BUSD) and wrapped tokens (WBTC). Membership never changes with hindsight.

A coin leaves the cross-section only when its price history ends. Of the 40, 39 have free price history; Revain (REV) has none and is the only drop.

Table 3. Universe: top 40 non-stablecoin coins by market cap on 2021-01-03.

Rank	Coin	Name	Source	First price	Last price	Status
1	BTC	Bitcoin	coinbase	2021-01-01	2026-10-08	active
2	ETH	Ethereum	coinbase	2021-01-01	2026-10-08	active
4	LTC	Litecoin	coinbase	2021-01-01	2026-10-08	active
5	XRP	XRP	kucoin	2021-01-01	2026-10-08	active
6	DOT	Polkadot	kucoin	2021-01-01	2026-10-08	active
7	BCH	Bitcoin Cash	coinbase	2021-01-01	2026-10-08	active
8	ADA	Cardano	kucoin	2021-01-01	2026-10-08	active
9	BNB	Binance Coin	gate	2021-01-01	2026-10-08	active
10	LINK	Chainlink	coinbase	2021-01-01	2026-10-08	active
13	BSV	Bitcoin SV	kucoin	2021-01-01	2026-10-08	active
14	XLM	Stellar	coinbase	2021-01-01	2026-10-08	active
15	EOS	EOS	bitfinex	2021-01-01	2026-07-03	history ends (delisted / collapsed)
16	XMR	Monero	bitfinex	2021-01-01	2026-10-08	active
17	THETA	THETA	gate	2021-01-01	2026-10-08	active
18	TRX	TRON	gate	2021-01-01	2026-10-08	active
19	XEM	NEM	gate	2021-01-01	2026-10-08	active
20	VET	VeChain	kucoin	2021-01-01	2026-10-08	active
21	XTZ	Tezos	coinbase	2021-01-01	2026-10-08	active
22	CEL	Celsius	gate	2021-04-09	2026-10-08	active
23	UNI	Uniswap	coinbase	2021-01-01	2026-10-08	active
24	LEO	UNUS SED LEO	bitfinex	2021-01-01	2026-10-08	active
25	CRO	Crypto.com Coin	kucoin	2021-01-01	2026-10-08	active
26	DOGE	Dogecoin	gate	2021-01-01	2026-10-08	active
27	ATOM	Cosmos	coinbase	2021-01-01	2026-10-08	active
30	NEO	Neo	kucoin	2021-01-01	2026-10-08	active
31	AAVE	Aave	coinbase	2021-01-01	2026-10-08	active
32	SNX	Synthetix	coinbase	2021-01-01	2026-10-08	active
33	REV	Revain	-	-	-	no free price history found
34	HT	Huobi Token	gate	2021-01-01	2026-10-08	active
35	FIL	Filecoin	gate	2021-01-01	2026-10-08	active
36	DASH	Dash	coinbase	2021-01-01	2026-10-08	active
37	IOTA	IOTA	bitfinex	2021-01-01	2026-10-08	active
38	ETC	Ethereum Classic	gate	2021-01-01	2026-10-08	active
39	ZIL	Zilliqa	gate	2021-01-01	2026-10-08	active
40	YFI	yearn.finance	coinbase	2021-01-01	2026-10-08	active
41	MKR	Maker	coinbase	2021-01-01	2026-01-12	history ends (delisted / collapsed)
42	FTT	FTX Token	bitfinex	2021-01-01	2022-11-29	history ends (delisted / collapsed)
43	COMP	Compound	coinbase	2021-01-01	2026-10-08	active
44	ZEC	Zcash	coinbase	2021-01-01	2026-10-08	active
45	SUSHI	SushiSwap	gate	2021-01-01	2026-10-08	active

3.2 Prices and sources

For each coin, daily candles were requested from three free public APIs - Coinbase Exchange (USD), KuCoin and Gate.io (USDT) and Bitfinex (USD) - and the source with the most complete history from January 2021 was used, breaking ties by dollar volume. The same source supplies hourly candles where available. USDT is treated as USD. Binance is geo-restricted from the research environment (HTTP 451). Bitfinex keeps the history of delisted pairs, which is how FTX Token (FTT, collapsed November 2022) enters the sample. Implied volatility is the Deribit DVOL index.

Several coins trade thinly at their chosen venue (median daily dollar volume below $100,000 for XEM, CEL and FTT). Their prices are real but noisy, and a fund could not trade them at size; Section 11.3 discusses this. Prices are aligned to a daily calendar, gaps of up to three days are forward-filled, and simple close-to-close returns are used.

3.3 Investment themes

Table 4. Themes, assigned from each project's function rather than from returns.

Theme	Coins
Payments & Privacy	BTC, LTC, BCH, BSV, XRP, XLM, DOGE, DASH, ZEC, XMR, XEM
Smart-Contract Platforms	ETH, ADA, DOT, EOS, TRX, XTZ, ATOM, NEO, ETC, ZIL
DeFi & Oracle	UNI, AAVE, SNX, YFI, MKR, COMP, SUSHI, LINK
Exchange & CeFi Tokens	BNB, LEO, CRO, HT, FTT, CEL
Data, IoT & Media	THETA, FIL, VET, IOTA

4. Backtest Methodology

4.1 Unconstrained backtest

Each day a signal is turned into weights using information up to that day's close; the weights earn the next day's return, so there is no look-ahead. Gross exposure (sum of absolute weights) is one. Cross-sectional strategies are dollar-neutral, with weights proportional to each coin's demeaned rank. Time-series strategies hold each coin long or short on its own signal, scaled by the inverse of its 30-day volatility. src/minimal_backtest.py reproduces this in about 60 lines and checks itself against the engine.

4.2 Costs

Costs follow the brief: 20 bps per unit of turnover for market orders and 7 bps for limit orders. Momentum-direction strategies are charged 20 bps because they chase price and need immediate fills. Reversal-direction strategies are charged 7 bps because they buy what has fallen and sell what has risen, supplying liquidity, and they trade slowly enough to work passive orders. Every result is also shown with all trades at 20 bps.

4.3 Metrics

Following the class convention, Sharpe = mean / standard deviation x sqrt(252) on daily returns, and annual return, volatility and alpha use 252 days. Crypto trades 365 days a year, so these are a common yardstick across projects rather than calendar-year figures (multiply by sqrt(365/252) = 1.20 for the calendar Sharpe). Alpha, beta and correlation come from daily returns against Bitcoin; the alpha t-statistic uses Newey-West (5-lag) standard errors. Maximum drawdown and CAGR use actual compounded returns.

4.4 Pre-registered protocol

The research period is 2021-07-01 to 2023-12-31; the hold-out is 2024-01-01 onward. The protocol, written before the hold-out was examined, makes every decision mechanically from research-period data:

- **Signals:** 51 signal variants across seven families - cross-sectional return (1-120 days, with and without a 14-day skip and volume weighting), time-series return (with and without a DVOL filter), theme return, residual versus the market, residual versus the coin's theme, short-term reversal in high-volatility regimes, and cointegrated pairs.

- **Direction:** each signal's direction - follow it (momentum) or fade it (reversal) - is the sign of its research-period gross Sharpe. Nothing about direction is chosen by hand.
- **Candidates:** every signal is crossed with a grid of holding periods, giving 167 candidates, each charged its book's cost.
- **Books:** in each book (momentum, reversal) the best variant of each family is kept if its research-period net Sharpe is positive; the book is an equal-weight ensemble of the top three families.
- **Execution and weighting:** the execution rule (holding period or no-trade band) for each book and the weighting method for combining the books are chosen by research-period Sharpe.

Exploration tables in Sections 5-6 show both periods for transparency, but only the research-period columns feed any decision. One caveat remains: version 1 results on a different 16-coin universe were seen before this protocol was written, and the protocol's ingredients (for example the DVOL filter and skip-period signals) were informed by that work. Because version 2 selects among them mechanically on 2021-2023 data, that influence is limited to which ingredients were available, not which were chosen.

5. Momentum Research (Research Period)

5.1 Time horizon

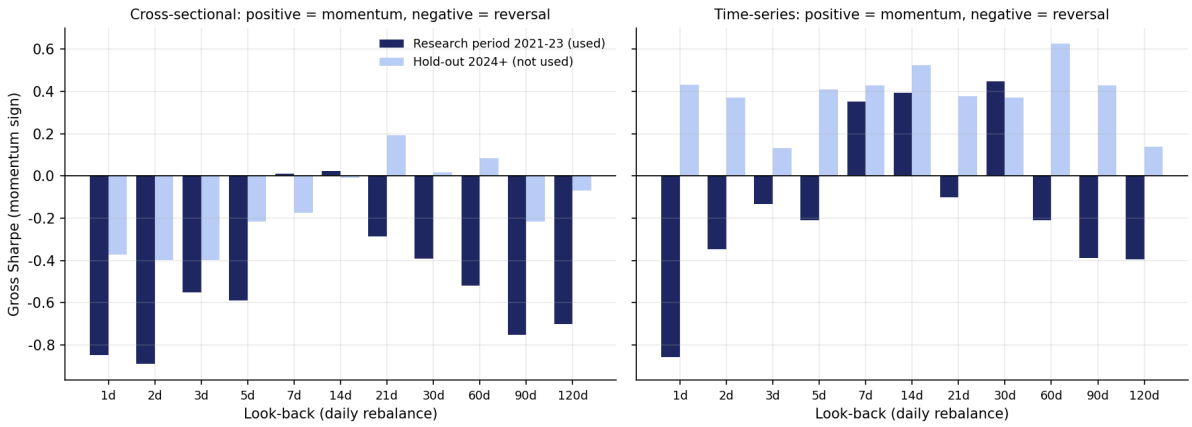


Figure 1. Gross Sharpe of momentum-sign strategies by look-back. Dark bars (research period) drive decisions; light bars (hold-out) are shown only for transparency.

In the research period the cross-sectional signal is reversal at 1-5 days (gross Sharpe -0.85 at 1 day, -0.59 at 5 days), flat at 7-14 days, and reversal again at 21-120 days (-0.75 at 90 days). Time-series momentum is positive at 7, 14 and 30 days (0.35 at 7 days, 0.39 at 14, 0.45 at 30; 21 days is slightly negative at -0.10) and turns to reversal at 60-120 days. With a broad survivorship-free universe, the brief's equity intuition - reversal at short horizons, momentum at medium ones - holds for the cross-section in crypto too, unlike the 16-coin result in version 1.

Table 5. Horizon scan (momentum sign; negative = reversal). IS = research period, OOS = hold-out.

Look-back	CS gross IS	CS gross OOS	CS net 7 IS	TS gross IS	TS gross OOS	TS net 20 IS	CS turnover
1 d	-0.85	-0.37	-2.36	-0.86	0.43	-2.12	1.33
3 d	-0.55	-0.40	-1.42	-0.13	0.13	-0.84	0.77
7 d	0.01	-0.17	-0.56	0.35	0.43	-0.12	0.51
14 d	0.02	-0.01	-0.39	0.39	0.52	0.07	0.37
30 d	-0.39	0.02	-0.67	0.45	0.37	0.23	0.26
60 d	-0.52	0.08	-0.73	-0.21	0.63	-0.39	0.19

90 d	-0.75	-0.21	-0.94	-0.39	0.43	-0.53	0.16
120 d	-0.70	-0.07	-0.87	-0.39	0.14	-0.52	0.13

5.2 New information and activity

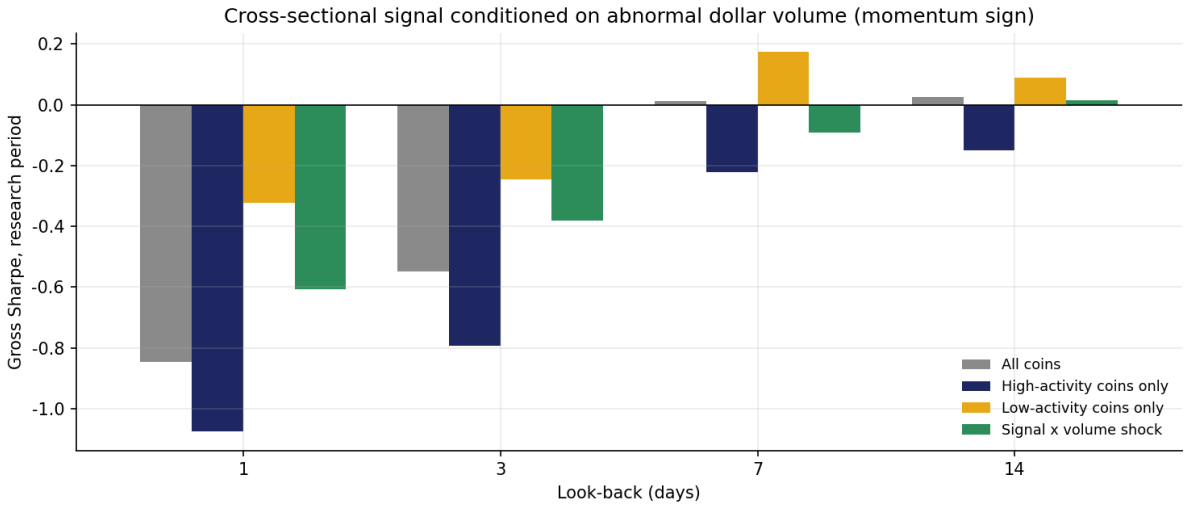


Figure 2. Cross-sectional signal conditioned on abnormal dollar volume, research period.

The brief suggests activity strengthens momentum and that uninformed (low-activity) moves reverse more. At 1-3 days the data show the reverse of the uninformed-trading story: reversal is stronger among high-activity coins (-1.07 at 1 day) than low-activity coins (-0.32). Volume shocks in this universe often mark liquidity-driven spikes in small coins that then retrace. Weighting by volume shock moves the signal toward momentum (1 day: -0.85 to -0.61), consistent with the brief's momentum hypothesis, but not enough to make a positive momentum signal.

5.3 Seasonality

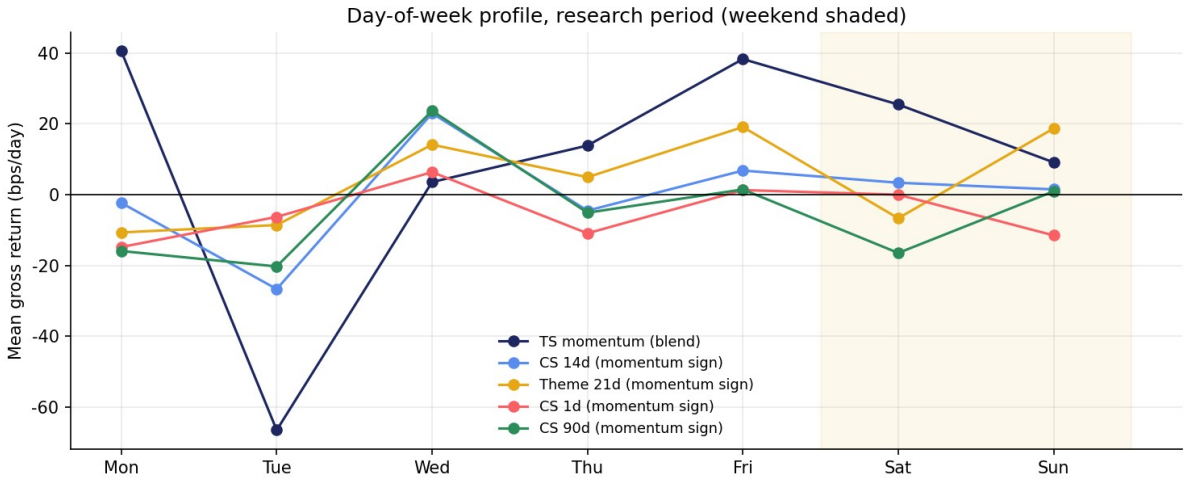


Figure 3. Day-of-week profile of strategy returns, research period (weekend shaded).

Time-series momentum is much stronger on weekends (gross Sharpe 1.37) than on weekdays (0.32), as in version 1: trends set by retail flow persist when institutions are away. Tuesday is consistently the worst day for trend-following.

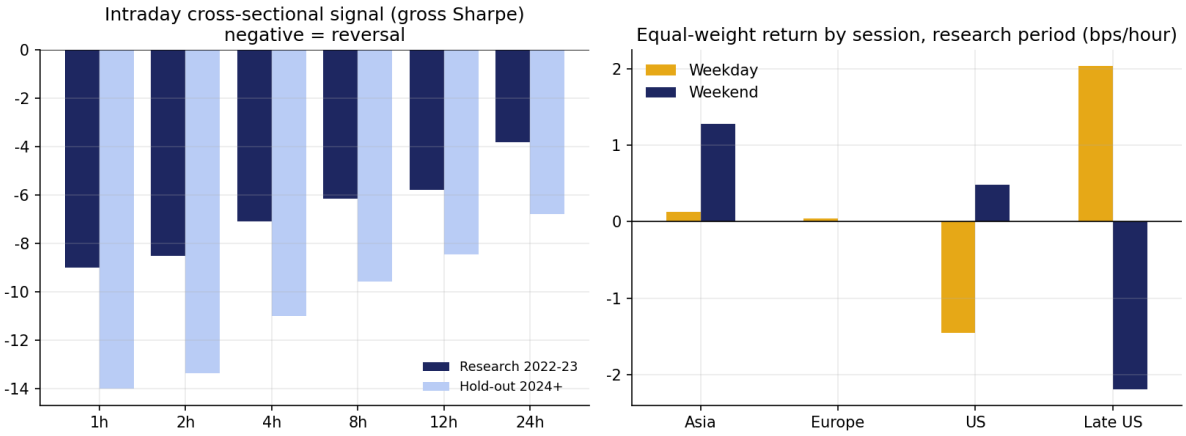


Figure 4. Left: intraday cross-sectional signal. Right: mean return by session.

Hourly data show the session pattern: on research-period weekdays the US session averaged -1.45 bps per hour and the late-US hours 2.04 bps per hour.

5.4 Investment themes

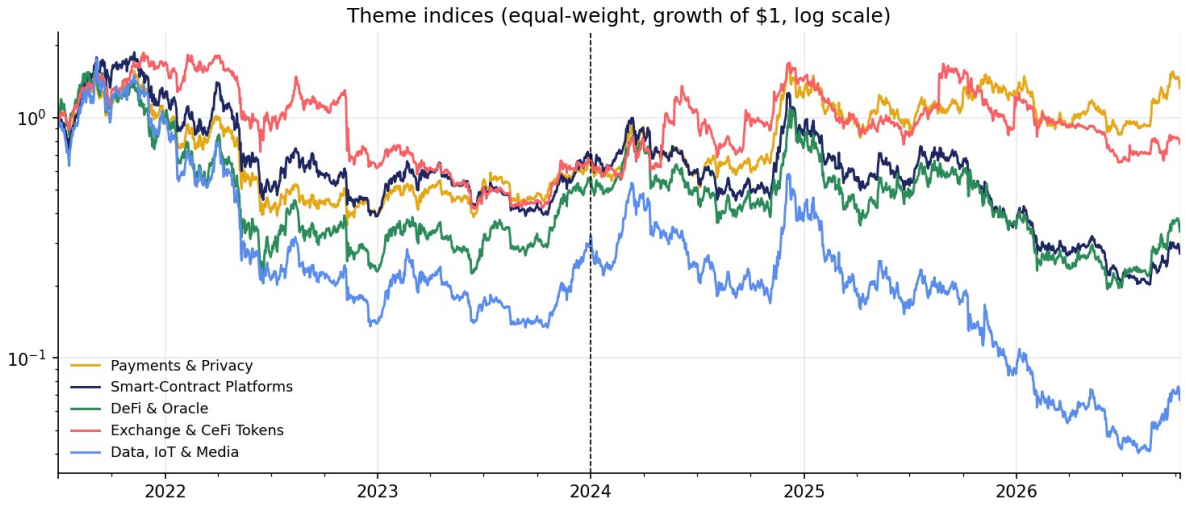


Figure 5. Equal-weight theme indices (growth of $1, log scale). Dashed line: hold-out start.

Theme momentum is positive at every look-back in the research period, best at 14 days (gross Sharpe 1.10 with a 7-day hold, net 0.57 at 20 bps). It was the top momentum candidate but decayed in the hold-out (gross 0.10).

5.5 Technical plays: front-running the US session

Table 6. Session-based front-running tests (hourly data, 2022 onward).

Test	Type	Gross bps/day	Gross IS	Gross OOS	Net at 7 bps	t-stat

Pre-US return -> US session	Time-series	-3.87	-0.96	0.07	-1.69	-0.93
Pre-US return -> US session	Cross-sectional	-2.98	-0.71	-0.67	-3.89	-1.73
Prior-day US session -> today's US session	Time-series	-5.70	-1.12	-0.09	-1.77	-1.39
Prior-day US session -> today's US session	Cross-sectional	0.14	0.09	-0.01	-3.17	0.09
US session -> late-US hours	Time-series	-1.12	0.36	-0.62	-2.46	-0.49
US session -> late-US hours	Cross-sectional	-4.96	-1.44	-2.35	-7.57	-5.10

Each session strategy opens and closes a full position every day, so even with limit orders the round trip costs 14 bps a day - far more than any of these effects earn. Front-running institutional hours is not profitable at the brief's costs.

6. Reversal Research (Research Period)

6.1 Time horizon

Daily cross-sectional reversal is strongest at 1-2 days (gross Sharpe 0.85 and 0.89 as reversal), but the 1-day version turns over 1.33x its capital a day and loses money even at 7 bps (net Sharpe -2.36). Slow reversal at 60-120 days trades a tenth as much.

Table 7. Intraday cross-sectional signal (momentum sign; negative = reversal), hourly rebalance.

Look-back	Gross IS	Gross OOS	Gross bps/hour	Turnover/hour	Net at 7 bps
1 h	-8.99	-14.00	-3.43	1.36	-43.6
2 h	-8.52	-13.36	-3.28	0.96	-34.1
4 h	-7.09	-11.02	-2.72	0.70	-26.2
8 h	-6.15	-9.58	-2.34	0.52	-20.8
12 h	-5.80	-8.46	-2.11	0.43	-17.9
24 h	-3.81	-6.78	-1.62	0.32	-13.3

Intraday, reversal is the dominant pattern in both periods, but two warnings apply. First, much of it is probably bid-ask bounce: in thinly traded coins the hourly close alternates between bid and ask prices, which looks like reversal but cannot be captured. Second, turnover of more than one times capital per hour puts it far out of reach at 7-20 bps. It is reported as evidence for the brief's horizon hypothesis, not as a strategy.

6.2 Correlation: residual baskets and cointegrated pairs

Table 8. Correlation-based signals. Negative gross = the residual reverses.

Strategy (momentum sign)	Gross IS	Gross OOS	Net 7 IS	Turnover
Residual vs market L=1d	-0.83	-0.55	-2.48	1.327
Residual vs theme L=1d	-1.19	-0.52	-2.87	1.337
Residual vs market L=3d	-0.70	-0.03	-1.70	0.765
Residual vs theme L=3d	-1.01	-0.10	-1.97	0.767
Residual vs market L=7d	-0.36	0.09	-1.01	0.513
Residual vs theme L=7d	-0.40	-0.14	-1.04	0.510
Residual vs market L=14d	-0.34	0.22	-0.83	0.370
Residual vs theme L=14d	-0.54	-0.01	-1.01	0.368
Residual vs market L=30d	-0.67	0.06	-0.99	0.255
Residual vs theme L=30d	-0.78	0.04	-1.09	0.254
Residual vs market L=60d	-0.55	0.18	-0.78	0.181
Residual vs theme L=60d	-0.76	-0.15	-1.00	0.185
Cointegrated pairs (Kalman, walk-forward)	-0.46	1.10	-0.58	0.038

'Security A minus something correlated to it' reverses at every horizon in the research period, against the market and even more against the coin's own theme (60-day theme residual -0.76 in momentum sign, i.e. 0.76 as reversal). Cointegrated pairs, traded with a Kalman-filter hedge ratio and monthly re-selection across 430 pairs, lost money in the research period (gross -0.46) and so were not selected - although they did well in the hold-out (1.10), a reminder of how unstable these relationships are.

6.3 Macro: volatility and dislocation

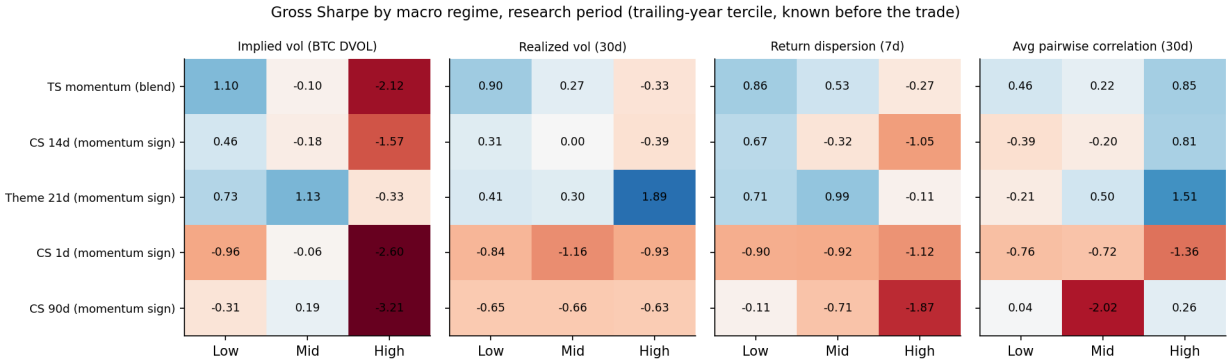


Figure 6. Gross Sharpe by macro regime, research period.

The clearest macro result is for momentum: time-series momentum earns 1.10 when implied volatility is low and -2.12 when it is high (the pattern repeats in the hold-out), which is why the DVOL filter is part of the selected strategy. For reversal, 1-day reversal is strongest when implied volatility is high (momentum-sign Sharpe -2.60) and when average correlation is high (-1.36), in line with the brief and Nagel (2012).

7. Mechanical Strategy Selection

Of the 51 signal variants, the research period assigned 17 to the momentum direction and 34 to reversal. After crossing with holding periods, the best candidate per family was:

Table 9. Family winners (research-period net Sharpe at the book's cost).

Best candidate in family	Book	Net Sharpe IS (selection)	Hold-out net 20	Hold-out net 7	Turnover
Theme look-back 14 momentum, hold 7	Momentum	0.57	-0.45	-0.09	0.17
TS look-back 7/14/28 blend + DVOL filter momentum, hold 1	Momentum	0.55	0.62	0.90	0.24
CS look-back 14 momentum, hold 3	Momentum	-0.56	-0.20	0.23	0.23
Residual vs theme look-back 60 reversal, hold 7	Reversal	0.81	-0.09	0.05	0.07
CS look-back 120 reversal, hold 7	Reversal	0.67	-0.16	-0.07	0.05
Residual vs market look-back 60 reversal, hold 7	Reversal	0.65	-0.40	-0.26	0.07
TS look-back 90 reversal, hold 1	Reversal	0.34	-0.58	-0.48	0.12
Short-term (1d) in high-vol regime reversal, hold 2	Reversal	0.29	-1.76	-0.66	0.27
Cointegrated pairs REV	Reversal	-0.58	0.64	0.94	0.04

Selected books (families with positive research-period net Sharpe, top three per book):

- **Momentum:** Theme look-back 14 momentum, hold 7
- **Momentum:** TS look-back 7/14/28 blend + DVOL filter momentum, hold 1
- **Reversal:** Residual vs theme look-back 60 reversal, hold 7
- **Reversal:** CS look-back 120 reversal, hold 7
- **Reversal:** Residual vs market look-back 60 reversal, hold 7

Cross-sectional momentum has no variant with a positive net Sharpe after 20 bps, so the momentum book has two members.

8. Execution

Table 10. Execution variants (book cost: momentum 20 bps, reversal 7 bps; 'other' swaps them).

Book	Execution	Turnover/ day	Gross Sharpe	IS at book cost	Full at book cost	Full at other cost
Momentum	Daily rebalance	0.206	0.98	0.65	0.48	0.81
Momentum	Hold 3d (overlapping)	0.180	0.81	0.46	0.39	0.66
Momentum	Hold 7d (overlapping)	0.146	0.64	0.25	0.32	0.53
Momentum	No-trade band 2%	0.163	0.95	0.75	0.54	0.81
Momentum	No-trade band 5%	0.043	-0.11	-0.40	-0.30	-0.18
Reversal	Daily rebalance	0.063	0.38	0.80	0.30	0.15
Reversal	Hold 3d (overlapping)	0.057	0.41	0.84	0.33	0.20
Reversal	Hold 7d (overlapping)	0.050	0.42	0.82	0.35	0.23
Reversal	No-trade band 2%	0.029	0.45	0.80	0.40	0.33
Reversal	No-trade band 5%	0.010	0.23	0.77	0.21	0.18

The research period selected 'No-trade band 2%' for momentum and 'Hold 3d (overlapping)' for reversal. A 2% band cuts momentum turnover without diluting the signal; wider bands or longer holds lose too much signal. The cost sweep below shows the combined portfolio reaches zero Sharpe between 20 and 30 bps in the hold-out (Sharpe 0.11 at 20 bps).

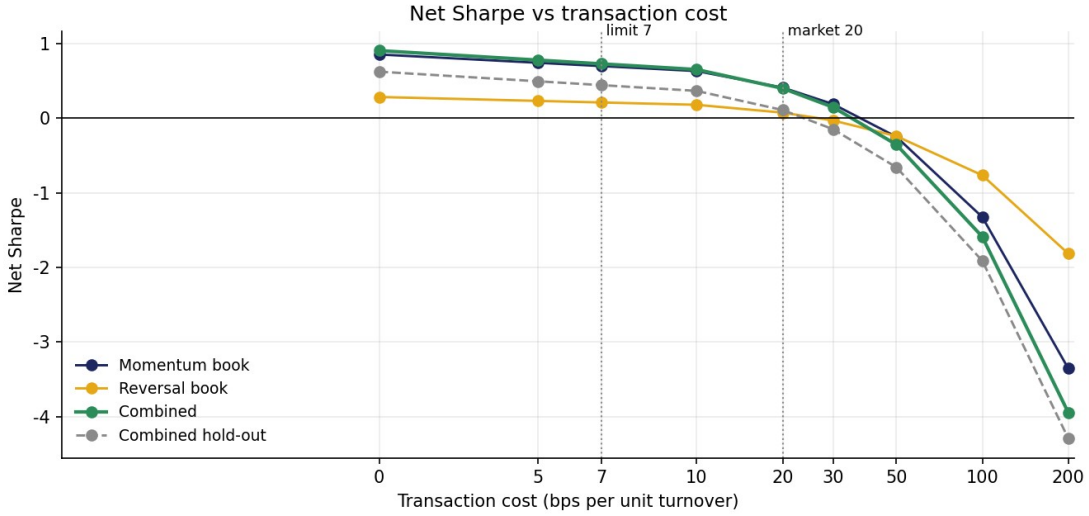


Figure 7. Net Sharpe versus transaction cost (0-200 bps).

9. Weighting

The two books are slightly negatively correlated (daily correlation -0.11). Four schemes were tested, all re-estimated every 21 days on the trailing 180 days only.

Table 11. Weighting methods (selection on the research-period Sharpe).

Method	Avg weight momentum	Avg weight reversal	Sharpe IS	Sharpe OOS	Sharpe full
Equal weight	50%	50%	0.86	0.20	0.49

Inverse volatility	43%	57%	0.93	0.20	0.50
Risk parity (ERC)	43%	57%	0.93	0.20	0.50
Mean-variance (shrunk)	47%	53%	0.60	-0.01	0.22

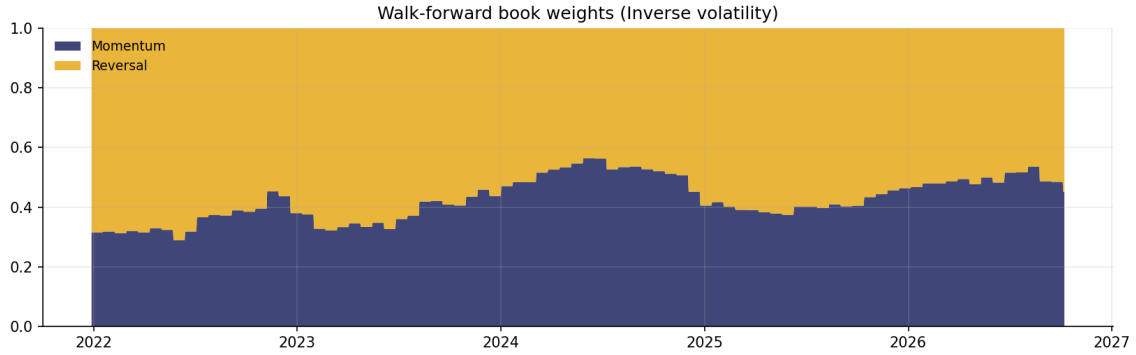


Figure 8. Walk-forward book weights (inverse volatility).

Table 12. Ensemble versus single best strategy.

Construction	Mom. IS	Mom. OOS	Rev. IS	Rev. OOS	Combined IS	Combined OOS
Final: top-3 family ensembles	0.54	0.30	0.66	-0.07	0.93	0.20
Single IS-best strategy per book	0.54	-0.30	0.70	0.08	0.98	-0.24

Picking only the single best research-period strategy in each book would have given a hold-out Sharpe of -0.24; the three-family ensembles gave 0.20. Diversifying across families reduces, but does not remove, selection risk.

10. Performance Evaluation

Table 13. Hold-out, 2024-01-01 to 2026-10-08 (never used for any decision).

Strategy	Ann. return	Ann. vol	Sharpe	Max DD	Alpha	Beta	Alpha t	Corr BTC
Combined portfolio *	**2.1%**	**10.3%**	**0.20**	**-26.7%**	**2.8%**	**-0.03**	**0.52**	**-0.12**
Combined (all trades at 20 bps)	1.1%	10.3%	0.11	-27.7%	1.9%	-0.03	0.34	-0.12
Momentum book (20 bps)	5.2%	17.3%	0.30	-24.8%	7.2%	-0.08	0.80	-0.18
Momentum book (7 bps)	10.6%	17.3%	0.61	-18.6%	12.5%	-0.08	1.40	-0.18
Reversal book (7 bps)	-1.0%	14.4%	-0.07	-35.4%	-1.2%	0.01	-0.16	0.03
Reversal book (20 bps)	-2.8%	14.4%	-0.19	-38.2%	-3.0%	0.01	-0.40	0.03
Cointegrated pairs (20 bps)	2.7%	4.3%	0.64	-5.6%	2.6%	0.01	1.20	0.06
Equal-weight crypto	10.1%	52.5%	0.19	-74.4%	-15.2%	1.05	-0.92	0.79
Buy & hold BTC	24.1%	39.4%	0.61	-53.1%	0.0%	1.00	0.33	1.00

** Momentum book at 20 bps (market orders), reversal book at 7 bps (limit orders). Sharpe = mean / std x sqrt(252); return, volatility and alpha annualised with 252 days. Alpha, beta and correlation vs daily Bitcoin returns; alpha t uses Newey-West (5-lag) errors.*

Table 14. Research period, 2021-12-28 to 2023-12-31 (all decisions made here).

Strategy	Ann. return	Ann. vol	Sharpe	Max DD	Alpha	Beta	Alpha t	Corr BTC
Combined portfolio *	**9.4%**	**10.1%**	**0.93**	**-13.3%**	**9.6%**	**-0.03**	**1.64**	**-0.15**
Combined (all trades at 20 bps)	8.2%	10.1%	0.81	-14.1%	8.3%	-0.03	1.43	-0.15
Momentum book (20 bps)	11.2%	20.6%	0.54	-17.1%	11.7%	-0.12	1.08	-0.28

Momentum book (7 bps)	16.6%	20.6%	0.81	-14.9%	17.1%	-0.12	1.59	-0.28
Reversal book (7 bps)	8.2%	12.5%	0.66	-16.5%	8.2%	0.01	1.18	0.04
Reversal book (20 bps)	6.3%	12.5%	0.50	-17.8%	6.2%	0.01	0.90	0.04
Cointegrated pairs (20 bps)	-5.3%	5.9%	-0.90	-14.9%	-5.4%	0.01	-1.68	0.09
Equal-weight crypto	-7.2%	55.8%	-0.13	-71.2%	-11.7%	1.05	-0.68	0.86
Buy & hold BTC	4.3%	45.8%	0.09	-67.0%	0.0%	1.00	3.65	1.00

Table 15. Full backtest, 2021-12-28 to 2026-10-08.

Strategy	Ann. return	Ann. vol	Sharpe	Max DD	Alpha	Beta	Alpha t	Corr BTC
Combined portfolio *	**5.2%**	**10.2%**	**0.50**	**-26.7%**	**5.7%**	**-0.03**	**1.42**	**-0.13**
Combined (all trades at 20 bps)	4.1%	10.2%	0.40	-27.7%	4.6%	-0.03	1.15	-0.13
Momentum book (20 bps)	7.7%	18.8%	0.41	-24.8%	9.3%	-0.10	1.34	-0.23
Momentum book (7 bps)	13.1%	18.7%	0.70	-18.6%	14.7%	-0.10	2.12	-0.23
Reversal book (7 bps)	2.9%	13.6%	0.21	-36.1%	2.7%	0.01	0.52	0.03
Reversal book (20 bps)	1.0%	13.6%	0.08	-40.2%	0.9%	0.01	0.16	0.03
Cointegrated pairs (20 bps)	-0.7%	5.0%	-0.13	-19.8%	-0.8%	0.01	-0.43	0.08
Equal-weight crypto	2.8%	53.9%	0.05	-74.4%	-13.8%	1.05	-1.15	0.82
Buy & hold BTC	15.8%	42.2%	0.37	-67.0%	0.0%	1.00	-1.36	1.00

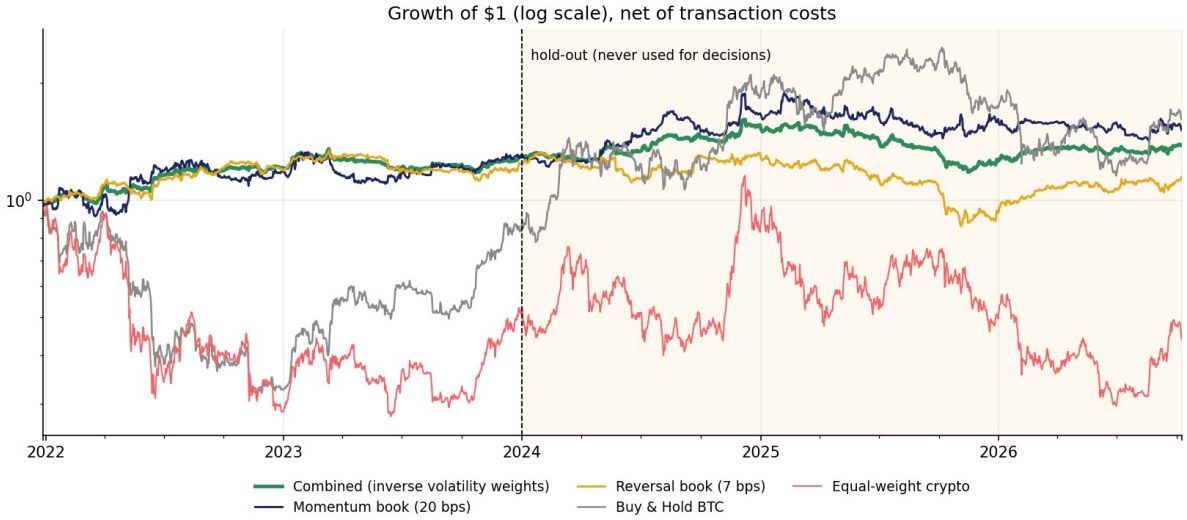


Figure 9. Growth of $1, net of costs (log scale). Shaded: hold-out.

In the hold-out the combined portfolio returned 2.1% a year at 10.3% volatility (Sharpe 0.20), with a maximum drawdown of -26.7% against -53.1% for Bitcoin. It is genuinely market-neutral - beta -0.03, correlation -0.12 - but its alpha (2.8%, t = 0.52) is not statistically distinguishable from zero. The momentum book held up (hold-out Sharpe 0.30 at 20 bps, 0.61 at 7 bps); the reversal book did not (-0.07).

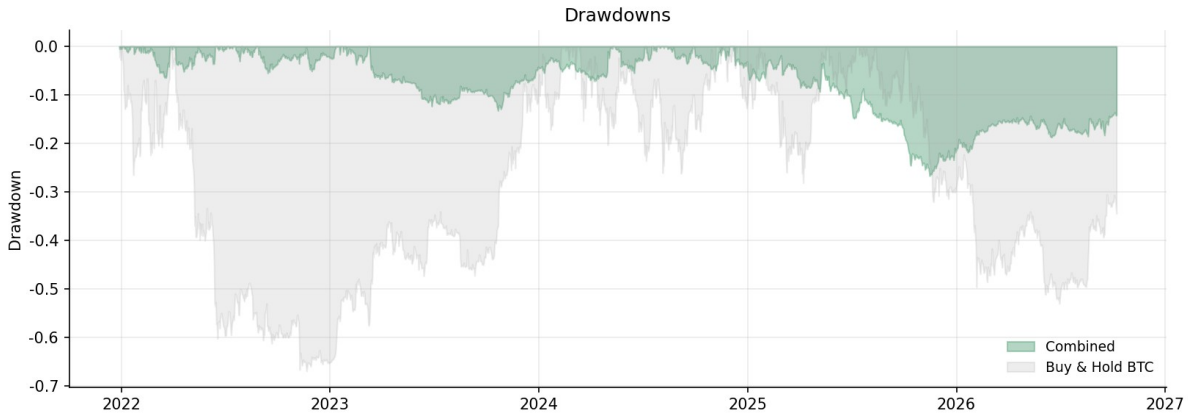


Figure 10. Drawdowns of the combined portfolio and Bitcoin.

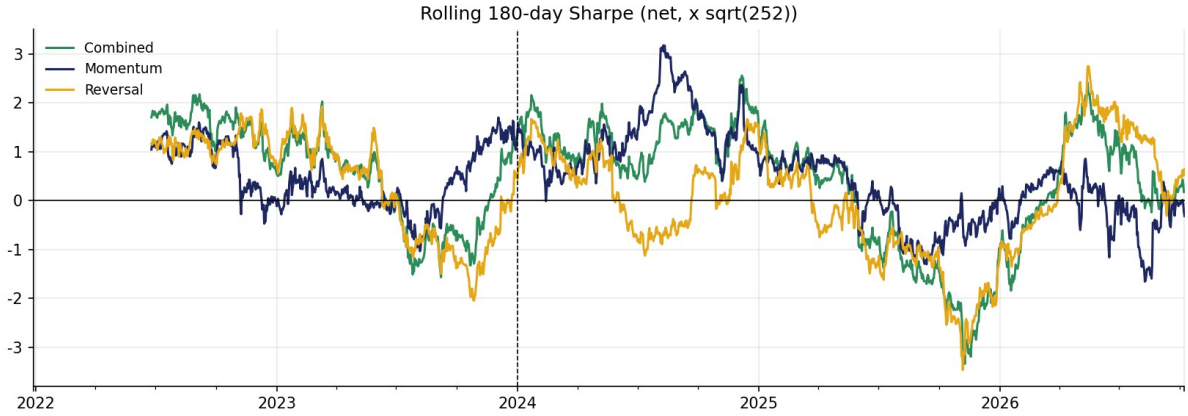


Figure 11. Rolling 180-day Sharpe ratio (net, x sqrt(252)). Dashed line: hold-out start.

11. Discussion

11.1 Version 1 versus version 2

Table 16. What the trainer's corrections did to the headline numbers.

	Version 1	Version 1 at sqrt(252)	Version 2
Universe	16 coins alive in 2026	16 coins alive in 2026	39 coins by market cap, Jan-2021
Decisions made on	Full sample (exploration)	Full sample (exploration)	2021-2023 only, mechanical
Combined Sharpe, full	1.02	0.85	0.50
Combined Sharpe, hold-out	0.55	0.46	0.20
Alpha t, hold-out	0.90	0.90	0.52
Correlation to BTC, full	-0.05	-0.05	-0.13

The sqrt(252) convention alone lowers every Sharpe by 17%. The larger change comes from the universe and the protocol: version 1's long-horizon reversal book was chosen partly because it worked in 2024-2026, and its 16 survivors excluded the coins whose collapses make cross-sectional strategies harder. Version 2's lower number is the more credible one.

11.2 What the research shows

- **Horizon.** Cross-sectional crypto returns reverse at short and long horizons; time-series returns trend at 1-4 weeks. Only the slower signals survive realistic costs.
- **The robust finding.** Time-series momentum with a DVOL filter was selected purely on 2021-2023 data and kept working in 2024-2026. It is the one finding that passed a genuine out-of-sample test.
- **Instability.** Theme momentum and slow residual reversal looked strong in 2021-2023 and faded. Pairs did the opposite. Relationships in crypto are unstable.
- **Bottom line.** The final portfolio is market-neutral and diversifying (correlation to BTC near zero), but its hold-out alpha is not statistically significant. It is a research result, not a deployable strategy.

11.3 Limitations

- **Shorting.** Short positions assume coins can be borrowed or shorted through perpetual futures; funding is not modelled.
- **Liquidity.** Several coins are thinly traded at their data venue; costs are linear and limit-order fills are assumed, which is optimistic for small coins and large capital.
- **Data.** Prices come from four venues (Coinbase, KuCoin, Gate.io, Bitfinex); venue differences add noise, and Revain is missing.
- **Multiple testing.** 167 candidates were compared; the research-period Sharpe of the winners is biased upward, which is why the hold-out leads.
- **Intraday coverage.** Hourly data are complete for 35 of 39 coins; CEL, HT and XEM have hourly prices only from September 2025 and FTT's hourly series is sparse, so the intraday cross-section is slightly smaller in 2022-2023.

11.4 Next steps

- Paper-trade the time-series momentum book with the DVOL filter on live data.
- Add perpetual-futures funding costs and a liquidity screen (minimum dollar volume) to the universe.
- Re-run the protocol each year as the hold-out grows, without changing its rules.

12. Conclusion

Applying the trainer's corrections - the sqrt(252) convention, a survivorship-free universe of 39 coins and a research protocol that never touches the hold-out - turns version 1's strong result into a modest one. Crypto shows reversal across the cross-section and momentum in individual coins' own trends, and the brief's conditioning variables (activity, seasonality, themes and especially implied volatility) all leave visible marks. But after costs, the combined portfolio's hold-out Sharpe is 0.20 with an alpha t-statistic of 0.52. The one component that survived a genuine out-of-sample test is time-series momentum switched off when implied volatility is high. The main lesson is methodological: how a strategy is selected matters as much as what it is.

Appendix A. Reproducibility

- python3 src/fetch_data.py - universe, daily, hourly and DVOL data
- python3 src/statarb.py - cointegrated pairs engine
- python3 src/research.py - research blocks, mechanical selection, weighting, performance

- python3 src/minimal_backtest.py - 60-line rebuild of one strategy, checked against the engine
- python3 src/make_report.py, build_explainer.py, build_pptx.py, build_excel.py, build_notebook.py

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